

Year 8 Mathematics

Mini -Test 4 - Circles, Rates, Ratio, Percentages

Resource Section (Calculators & notes allowed)



Time: 30 mins Name: Solutions

/30 marks

Questions 1 to 5 are Multiple Choice

Circle

[1 mark each]

1. Find the rate in dollars per kilogram for 800g of apples that cost \$3.00
 a) \$2.75/kg b) \$3.75/kg c) \$4.75/kg d) \$2.66/kg
2. Find 15% off a \$33 hair cut:
 a) \$3.30 b) \$29.70 c) \$4.95 d) \$28.05
3. Divide \$60 in the ratio of 4:2
 a) \$20: \$40 b) \$40: \$20 c) \$6: \$10 d) \$8: \$12
4. Can you simplify the ratio 6:16 ?
a) 3: 8 b) 1:11 c) No, it's not possible d) 2:22
5. If Brad travels at 80km/h on his moped for 4hrs, how far will he go?
 a) 20km b) 80km c) 240km d) 320km

Questions 6 to 11 are not multiple choice

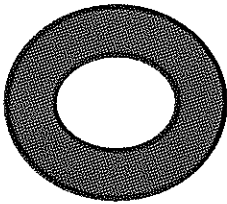
6. Convert each of the following units. [6 Marks]
 a) 125 cm = 1.25 [m] b) 185 mm = 18.5 [cm] c) 0.05g = 50 [mg]
 d) 140 kl = 0.14 [L] e) 13m² = 130,000 [cm²] f) 152 mm³ = 0.152 [cm³]
7. A hotel room costs \$145 per night normally but is "**seasonally adjusted**". [4 marks]
 a) Increase it by 10%.... b) Decrease it by 15%.....

\$159.50 ✓
 x 1.10
 or find 10% + add it

\$123.25 ✓

15

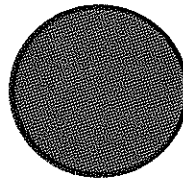
8. Calculate the circumference of each of these circles (2 decimal places). [3 marks]



a) radius: 6.9 cm

diameter = $\checkmark$
13.8 cm

43.35 cm $\checkmark$

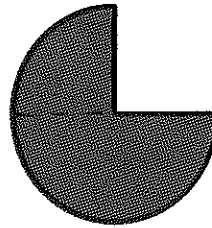
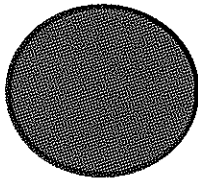


b) diameter 43 cm

135.09 cm $\checkmark$

c) Find the area of the circle & part circle. [2 marks]

Both have the same radius 2.6 m



9. Which is the best buy 200g for \$5:00 or 500g for \$10? [2 Marks]

100g \$2.50
1kg \$25
100g \$2.00
1kg \$20

10. (a) Find the profit made on a 2-bedroom unit bought for \$670 000 5 years ago, that just sold for \$735 000. [2 marks]

(1 dec place)

$\frac{\$65,000}{\$670,000} \checkmark$
9.7% $\checkmark$

(b) Find the loss on a 4 x 2 x 2 home bought for \$715 000 6 years ago, that just sold for \$695 000. [2 marks]

(1 dec place)

$\frac{\$20,000}{\$715,000} \checkmark$
2.79% $\checkmark$

11. A belt is placed around the equator... [4 marks]

a) What would be the length of the belt assuming a diameter of 12800 km?

$\pi \times d = C$
 $C = 402123.86 \text{ km}$ (2 decpl)

b) if 10m was added to the length of the belt, what would be the distance between the belt and the Earth's surface?

3.18 km

12800 $\div$ 3.18 km

-END of TEST-



402133.86

402123.86

Extension Qns: (10) What was the average yearly profit & loss for these property investments?

$\frac{9.7 + 2.8}{2} = 6.25\%$